General Dessert #12: Mobile and Portable HF Stations, Alternative Energy

Capacitance hats and corona balls on mobile whips, fused DC connections, vehicle-noise sources, short-antenna bandwidth limits, solar panels, charge controllers, and series blocking diodes.

Group G4E · 11 pool questions · Self-study reference · Practice exams at eham.net/exams

Key Facts Reference

Mobile HF Antennas

Component / topic	Answer	Q
Capacitance hat purpose	**Electrically lengthen a physically short antenna** (extends the effective current distribution along the element, increasing radiation resistance).	G4E01
Corona ball purpose	**Reduce RF voltage discharge from the tip of the antenna while transmitting.**	G4E02
Main limitation of HF mobile installations	**Efficiency of the electrically short antenna.** The whip is the bottleneck, not the rig.	G4E05
Disadvantage of a shortened mobile antenna vs. full-size	**Operating bandwidth may be very limited** (high Q narrows it).	G4E06

DC Power and Vehicle Interference

Question	Answer	Q
Best fused DC connection for a 100 W HF mobile	**Direct to the battery using heavy-gauge wire.**	G4E03
Why not the auxiliary (cigarette-lighter) socket?	**The socket's wiring may be inadequate for the current** drawn by the transceiver.	G4E04
Possible sources of receive interference in a vehicle	**All of the above**: charging system, fuel-delivery system, control computers.	G4E07

Solar Panels and Alternative Energy

Question	Answer	Q
Cell interconnection inside a solar panel	**Series-parallel.** Series for voltage, parallel sets for current.	G4E08
Open-circuit voltage of a fully illuminated silicon PV cell	**About 0.5 VDC.**	G4E09
Why use a series diode between panel and battery	**To prevent discharge of the battery back through the panel during low or no illumination.**	G4E10
Precaution for solar panel charging a LiFePO4 battery	**The solar panel must have a charge controller** – lithium chemistries need regulated charging.	G4E11

Dessert 12 Question Bank (G4E)

G4E01. What is the purpose of a capacitance hat on a mobile antenna?
A. To increase the power handling capacity of a whip antenna
B. To reduce radiation resistance
C. To electrically lengthen a physically short antenna
D. To lower the radiation angle

G4E02. What is the purpose of a corona ball on an HF mobile antenna?
A. To narrow the operating bandwidth of the antenna
B. To increase the “Q” of the antenna
C. To reduce the chance of damage if the antenna should strike an object
D. To reduce RF voltage discharge from the tip of the antenna while transmitting

G4E03. Which of the following direct, fused power connections would be the best for a 100-watt HF mobile installation?
A. To the battery using heavy-gauge wire
B. To the alternator or generator using heavy-gauge wire
C. To the battery using insulated heavy duty balanced transmission line
D. To the alternator or generator using insulated heavy duty balanced transmission line

G4E04. Why should DC power for a 100-watt HF transceiver not be supplied by a vehicle’s auxiliary power socket?
A. The socket is not wired with an RF-shielded power cable
B. The socket’s wiring may be inadequate for the current drawn by the transceiver
C. The DC polarity of the socket is reversed from the polarity of modern HF transceivers
D. Drawing more than 50 watts from this socket could cause the engine to overheat

G4E05. Which of the following most limits an HF mobile installation?
A. “Picket fencing”
B. The wire gauge of the DC power line to the transceiver
C. Efficiency of the electrically short antenna
D. FCC rules limiting mobile output power on the 75-meter band

G4E06. What is one disadvantage of using a shortened mobile antenna as opposed to a full-size antenna?
A. Short antennas are more likely to cause distortion of transmitted signals
B. Q of the antenna will be very low
C. Operating bandwidth may be very limited
D. Harmonic radiation may increase

G4E07. Which of the following may cause receive interference to an HF transceiver installed in a vehicle?
A. The battery charging system
B. The fuel delivery system
C. The control computers
D. All these choices are correct

G4E08. In what configuration are the individual cells in a solar panel connected together?
A. Series-parallel
B. Shunt
C. Bypass
D. Full-wave bridge

G4E09. What is the approximate open-circuit voltage from a fully illuminated silicon photovoltaic cell?
A. 0.02 VDC
B. 0.5 VDC
C. 0.2 VDC
D. 1.38 VDC

G4E10. Why should a series diode be connected between a solar panel and a storage battery that is being charged by the panel?
A. To prevent overload by regulating the charging voltage
B. To prevent discharge of the battery through the panel during times of low or no illumination
C. To limit the current flowing from the panel to a safe value
D. To prevent damage to the battery due to excessive voltage at high illumination levels

G4E11. What precaution should be taken when connecting a solar panel to a lithium iron phosphate battery?
A. Ground the solar panel outer metal framework
B. Ensure the battery is placed terminals-up
C. A series resistor must be in place
D. The solar panel must have a charge controller

Answer Key

G4E01: C	G4E02: D	G4E03: A	G4E04: B
G4E05: C	G4E06: C	G4E07: D	G4E08: A
G4E09: B	G4E10: B	G4E11: D	